Notes: Chen, Roll, and Ross (JB, 1986)
Economic Forces and the Stock Market

Contents

1 Big picture

- **Question.** Which macroeconomic innovations are systematic risk factors that are priced in the stock market?
- **Motivation.** Asset-pricing theory says that only systematic risks should earn risk premia, but it gives little direct guidance on the identity of the systematic state variables.
- **Core idea.** Stock prices are discounted claims on expected cash flows, so news that changes expected cash flows or discount rates should affect returns and may be priced.
- **Candidate forces.** The paper studies industrial production, expected and unexpected inflation, the risk-premium spread between low-grade and government bonds, the term-structure spread between long and short rates, market indices, consumption, and oil prices.
- **Data period.** The macro data run mainly from January 1953 to November 1983. The main cross-sectional pricing tests begin in 1958 because the paper uses the previous 60 months to estimate factor exposures.
- **Portfolio design.** Stocks are grouped into 20 equally weighted portfolios by firm size. The size sort gives dispersion in average returns and reduces noise and errors-in-variables problems in estimated betas.
- **Main result.** Industrial production, changes in the risk premium, twists in the yield curve, and inflation-related variables are significantly priced in stock returns.
- **Market-index result.** The value-weighted and equally weighted NYSE indices explain time-series return variation, but their betas do not explain cross-sectional expected returns once macro state-variable betas are included.
- **Consumption result.** Real consumption growth is not significantly priced and has the wrong sign in the paper's consumption-beta tests.
- **Oil result.** Oil price changes are not independently priced, even in the 1968-77 period when OPEC became important.
- **Economic takeaway.** Expected returns are better explained by exposure to identifiable macroeconomic news than by exposure to a stock market index alone.

2 Data and measurement

2.1 Why macroeconomic state variables matter

- The paper begins from the idea that asset prices respond to economic news.
- Theory emphasizes systematic risks, not idiosyncratic risks, because diversified investors do not require compensation for bearing diversifiable risk.
- The central empirical challenge is identifying the relevant state variables.
- Chen, Roll, and Ross choose variables using simple valuation logic rather than a purely statistical factor-extraction approach.

Interpretation. The goal is not to ask whether stocks move with macro variables in the time series only. The paper asks whether cross-sectional exposures to those macro variables command risk premia.

2.2 Discount rates and expected cash flows

The paper uses a simple valuation formula:

$$p = \frac{E(c)}{k}, \tag{1}$$

where p is price, $E(c)$ is the expected dividend stream, and k is the discount rate. The corresponding return intuition is

$$\frac{dp}{p} + \frac{c}{p} = \frac{d[E(c)]}{E(c)} - \frac{dk}{k} + \frac{c}{p}. \tag{2}$$

Interpretation. Systematic forces that alter expected cash flows or discount rates should alter stock returns. This motivates the selection of real activity, inflation, default-spread, and term-structure variables.

2.3 Industrial production

- Industrial production is the main real-activity measure.
- Monthly industrial production growth is

$$MP(t) = \log IP(t) - \log IP(t-1).$$

- Annual industrial production growth is

$$YP(t) = \log IP(t) - \log IP(t-12).$$

- Because industrial production data are released with a timing lag, the monthly production variable is led by one month in the pricing tests.

Interpretation. Industrial production is a cash-flow factor: higher expected real activity raises the value of firms' cash flows.

2.4 Inflation variables

- Monthly inflation $I(t)$ is the first difference in the logarithm of the Consumer Price Index.
- Expected inflation is based on the Fama-Gibbons procedure.
- Unexpected inflation is

$$UI(t) = I(t) - E[I(t) \mid t-1].$$

- Expected inflation is obtained using Fisher's equation:

$$TB(t-1) = E[RHO(t) \mid t-1] + E[I(t) \mid t-1],$$

  where TB is the one-month Treasury-bill rate and RHO is the real interest rate.
- The change in expected inflation is

$$DEI(t) = E[I(t+1) \mid t] - E[I(t) \mid t-1].$$

Interpretation. Inflation can affect both nominal cash flows and nominal discount rates. It can also redistribute wealth across assets with different nominal rigidity.

2.5 Risk premium and term structure

The paper defines two bond-market variables:

$$UPR(t) = Baa(t) - LGB(t), \tag{3}$$

where $Baa(t)$ is the return on low-grade corporate bonds and $LGB(t)$ is the return on long-term government bonds, and

$$UTS(t) = LGB(t) - TB(t-1). \tag{4}$$

- UPR is a proxy for an unanticipated change in the aggregate risk premium.
- UTS captures the term-structure effect, or the unanticipated return on long bonds relative to short bills.
- In the data, UPR and UTS are highly negatively correlated because both use long-term government bonds.

Interpretation. These variables are discount-rate factors. The risk premium captures changing compensation for bearing risk, while the term-structure variable captures shifts in long rates relative to short rates.

2.6 Market indices, consumption, and oil

- The market-index variables are:

  $EWNY(t)$ = return on the equally weighted NYSE index, $VWNY(t)$ = return on the value-weighted NYSE index.

- The consumption variable CG is real per capita consumption growth.
- The oil variable OG is the log change in the producer price index for crude petroleum.
- The market indices are included as competitors to the macro variables.
- Consumption and oil prices are included as natural alternative state variables.

Interpretation. A major contribution of the paper is that it compares the macro state variables directly against the stock market index, consumption, and oil prices.

2.7 Sample, subperiods, and portfolios

- The correlation analysis covers January 1953 to November 1983 and several subperiods.
- The cross-sectional pricing tests use monthly stock returns on 20 equally weighted portfolios.
- Portfolios are grouped by market capitalization at the beginning of each test period.
- Factor exposures are estimated using the previous 60 months of returns.
- The paper reports tests for the full period and subperiods 1958-67, 1968-77, and 1978-84.

Interpretation. Size-sorted portfolios give useful dispersion in average returns while reducing noise relative to individual stocks.

3 Models and empirical specifications

3.1 Economic state-variable return model

The main time-series model for portfolio returns is:

$$R = a + b_{MP}MP + b_{DEI}DEI + b_{UI}UI + b_{UPR}UPR + b_{UTS}UTS + e. \tag{5}$$

- R is the return on a size portfolio.
- The b coefficients are portfolio exposures to macroeconomic innovations.
- e is idiosyncratic variation.

Interpretation. The paper first estimates exposures to macro news and then asks whether those exposures explain average returns.

3.2 Fama-MacBeth pricing procedure

The empirical procedure is a version of Fama-MacBeth cross-sectional testing:

1. Choose a sample of assets: 20 size-sorted portfolios.
2. Estimate each portfolio's exposure to economic variables using a rolling 60-month time-series regression.
3. Use the estimated betas as independent variables in 12 monthly cross-sectional regressions over the following year.
4. Repeat this procedure year by year.
5. Test whether the time-series mean of each estimated risk premium differs from zero.

Interpretation. A macro factor is priced if portfolios with higher exposure to that factor earn different average returns.

3.3 Pricing equation

The factor-pricing restriction is

$$E - r = b^* \lambda, \tag{6}$$

where $E - r$ is the vector of excess returns, b is a vector or matrix of betas, and λ is the vector of risk premia.

Interpretation. The question is whether λ is significantly different from zero for macro variables.

3.4 Market-index tests

The paper uses two related tests.

First, it augments the macro-factor specification with an index:

$$E - r = b^*_{macro} \lambda_{macro} + b^*_M \lambda_M. \tag{7}$$

Second, it reverses the null and asks whether macro factors add explanatory power beyond market-index betas:

$$E - r = b^*_M \lambda_M + b^*_{macro} \lambda_{macro}. \tag{8}$$

Interpretation. If the CAPM index is sufficient, macro betas should not matter once the market beta is included. The paper finds the opposite.

3.5 Consumption-beta model

The consumption-based null is:

$$E - r = b^*_c k, \tag{9}$$

where b_c is the vector of consumption betas and k is the consumption risk premium. The alternative is:

$$E - r = b^*_c k + b^* q, \tag{10}$$

where b contains the macro-state betas and q contains their risk premia.

Interpretation. Under the consumption-beta null, k should be positive and the macro-state premia q should be zero. The data reject this simple consumption interpretation.

3.6 Oil-price test

The oil test adds oil growth to the macro-factor pricing equation:

$$E - r = b^*_{OG}\lambda_{OG} + b^*_{macro}\lambda_{macro}. \tag{11}$$

Interpretation. If oil-price shocks are a priced systematic risk, λ_{OG} should be significant. The paper finds no overall pricing role for oil.

4 Identification and empirical design

4.1 What identifies the pricing effects

- The paper uses cross-sectional differences in portfolio exposures to macroeconomic innovations.
- A factor is priced when portfolios with different betas earn different average returns.
- The rolling beta procedure gives time-varying estimates of exposure.
- The Fama-MacBeth approach estimates the average premium associated with each exposure.

Interpretation. The evidence is strongest when macro factors remain priced after the market index, consumption, and oil are included.

4.2 Why the variables are theory-guided

- The authors do not rely solely on statistical factor extraction.
- The variables are chosen because they plausibly affect either discount rates or expected cash flows.
- Industrial production captures real activity and cash-flow news.
- Inflation variables capture nominal and real valuation changes.
- Bond-market spreads capture risk-premium and term-structure news.

Interpretation. This is the paper's methodological contribution: it tries to put economic names on systematic risk factors.

4.3 Why the paper avoids a VAR innovation approach

- The authors note that a vector autoregression might use lagged stock returns to forecast macro variables.
- That would indirectly use stock-market information to explain stock returns.
- The paper therefore uses simple, theory-based variable construction.
- The cost is possible errors-in-variables bias when expected components are not fully removed.

Interpretation. The empirical design trades statistical sophistication for a cleaner interpretation of macro factors as non-equity state variables.

4.4 Why subperiods matter

- The authors split the sample around 1973 and 1978 because oil prices and inflation changed sharply in the 1970s.
- Inflation-related variables are strongest in 1968-77, when inflation was volatile.
- The later 1978-84 subperiod has fewer observations and weaker estimates.

Interpretation. The pricing of inflation risks appears state-dependent, but the real-activity and bond-spread variables are more stable.

4.5 What the paper cannot fully prove

- The macro variables may proxy for omitted state variables rather than being primitive sources of risk.
- The variables may contain expected components, generating measurement error.
- The portfolio sort by size spreads average returns but does not itself provide a theory of size.
- The results are pre-1984 and may not describe later financial markets.
- The paper is not a causal design. It is an asset-pricing test of whether exposures to economic news command risk premia.

Interpretation. The paper's conclusion is not that these are the only state variables, but that they are economically meaningful and outperform several famous alternatives.

5 Tables

5.1 Table 1 and Table 2: variables and correlations

Read: Table 1:

- The table defines the basic series: inflation, Treasury bills, long government bonds, industrial production, low-grade bonds, market indices, consumption, and oil prices.
- It also defines the derived series used in the pricing tests: MP, YP, expected inflation, UI, RHO, DEI, UPR, and UTS.
- The key macro-pricing variables are MP, DEI, UI, UPR, and UTS.

Read: Table 2:

- The value-weighted and equally weighted NYSE indices are highly correlated: 0.916 for the full sample.
- The strongest non-market correlation is between UPR and UTS: -0.752 in the full sample and -0.890 in 1978-83.
- Production growth is only weakly correlated with the market indices.
- The correlations are not so high as to make all macro variables redundant, although UPR and UTS are clearly collinear.

Economic message. Table 1 shows that the paper translates theory into concrete state variables. Table 2 shows that the variables are related but not identical, so the pricing tests can ask which risks are actually rewarded.

TABLE 1 **Glossary and Definitions of Variables**

Symbol	Variable	Definition or Source
	Basic Series	
I	Inflation	Log relative of U.S. Consumer Price Index
TB	Treasury-bill rate	End-of-period return on 1-month bills
LGB	Long-term government bonds	Return on long-term government bonds (1958–78: Ibbotson and Sinquefield [1982]; 1979–83: CRSP)
IP	Industrial production	Industrial production during month (*Survey of Current Business*)
Baa	Low-grade bonds	Return on bonds rated Baa and under (1953–77: Ibbotson [1979], constructed for 1978–83)
EWNY	Equally weighted equities	Return on equally weighted portfolio of NYSE-listed stocks (CRSP)
VWNY	Value-weighted equities	Return on a value-weighted portfolio of NYSE-listed stocks (CRSP)
CG	Consumption	Growth rate in real per capita consumption (Hansen and Singleton [1982]; *Survey of Current Business*)
OG	Oil prices	Log relative of Producer Price Index/Crude Petroleum series (Bureau of Labor Statistics)
	Derived Series	
MP(t)	Monthly growth, industrial production	$\log_e[\mathrm{IP}(t)/\mathrm{IP}(t-1)]$
YP(t)	Annual growth, industrial production	$\log_e[\mathrm{IP}(t)/\mathrm{IP}(t-12)]$
E[I(t)]	Expected inflation	Fama and Gibbons (1984)
UI(t)	Unexpected inflation	$\mathrm{I}(t) - \mathrm{E}[\mathrm{I}(t)\|t-1]$
RHO(t)	Real interest (ex post)	$\mathrm{TB}(t-1) - \mathrm{I}(t)$
DEI(t)	Change in expected inflation	$\mathrm{E}[\mathrm{I}(t+1)\|t] - \mathrm{E}[\mathrm{I}(t)\|t-1]$
URP(t)	Risk premium	$\mathrm{Baa}(t) - \mathrm{LGB}(t)$
UTS(t)	Term structure	$\mathrm{LGB}(t) - \mathrm{TB}(t-1)$

Table 1: Glossary and definitions

TABLE 2 **Correlation Matrices for Economic Variables**

Symbol	EWNY	VWNY	MP	DEI	UI	UPR	UTS
	A. January 1953–November 1983						
VWNY	.916						
MP	.103	.020					
DEI	−.163	−.119	.063				
UI	−.163	−.112	−.067	.378			
UPR	.105	.042	.216	.266	.018		
UTS	.227	.248	−.159	−.394	−.103	−.752	
YP	.270	.270	.139	−.003	−.005	.113	.099
	B. January 1953–December 1972						
VWNY	.930						
MP	.147	.081					
DEI	−.130	−.122	.020				
UI	−.081	−.021	−.203	.388			
UPR	.265	.214	.213	.068	−.072		
UTS	.110	.108	−.059	−.210	−.041	−.688	
YP	.260	.238	.128	−.013	−.032	.128	.063
	C. January 1973–December 1977						
VWNY	.883						
MP	.022	−.118					
DEI	−.314	−.263	.004				
UI	−.377	−.352	−.004	.505			
UPR	.341	.231	.227	.032	−.289		
UTS	.217	.313	−.350	−.280	.026	−.554	
YP	.335	.361	.107	−.124	−.334	.221	.174
	D. January 1978–November 1983						
VWNY	.937						
MP	.092	−.010					
DEI	−.143	−.073	.169				
UI	−.055	−.024	.168	.375			
UPR	−.275	−.319	.248	.458	.259		
UTS	.424	.431	−.277	−.512	−.239	−.890	
YP	.269	.261	.193	.053	.247	.018	.115

NOTE.—VWNY = return on the value-weighted NYSE index; EWNY = return on the equally weighted NYSE index; MP = monthly growth rate in industrial production; DEI = change in

Table 2: Correlation matrices

5.2 Table 3: autocorrelations of economic variables

Read:

- YP is highly autocorrelated, with a first-order autocorrelation of 0.9615 and a very high adjusted Box-Pierce statistic.
- MP has strong seasonal behavior; the 12-month autocorrelation is 0.8030.
- Many variables have mild autocorrelation, which means that imperfect filtering can bias estimated betas and lower statistical significance.
- Market indices are much less autocorrelated than the macro variables.

Economic message. The macro variables are not clean white-noise innovations. This makes the paper's results more conservative because measurement error in betas tends to reduce significance.

TABLE 3 **Autocorrelations of the Economic Variables, January 1953–November 1983**

Symbol	ρ_1	ρ_2	ρ_3	ρ_4	ρ_5	ρ_6	ρ_7	ρ_8	ρ_9	ρ_{10}	ρ_{11}	ρ_{12}	Adjusted Box/Pierce (24)
YP	.9615	.8896	.7937	.6838	.5658	.4477	.3290	.2088	.0919	−.0196	−.1233	−.2109	1,639
MP	−.0990	−.1711	−.1204	.0413	.0778	.0241	.0765	.0240	−.1558	−.2122	−.1914	.8030	632.9
DEI	−.0432	−.0864	−.0094	−.0719	.0284	.0130	−.0874	.1662	.1101	−.0290	.0297	.0007	43.33
UI	.1804	.1314	.0567	.0483	.0490	−.0454	−.0398	.0535	.1391	.1536	.1361	.1875	85.50
UPR	−.1053	.0491	−.1340	.0882	−.0196	.0422	−.1297	.0117	−.0494	−.0733	.0834	.0264	52.81
UTS	.0267	−.0052	−.1637	.0383	.0739	.0750	−.0929	−.0278	.0023	−.0105	.1693	−.0029	57.15
CG	−.2458	−.0269	.1190	.0192	−.0460	.0082	.0497	−.0496	.0121	.0470	.1364	−.1324	53.06
OG	.4088	.2194	.1523	.1613	.0954	.1447	.1594	.0674	.0969	.0976	.0609	−.0038	159.0
EWNY	.1447	−.0133	.0141	.0554	.0518	−.0213	−.0959	−.0861	.0072	−.0140	.0043	.0997	40.43
VWNY	.0677	−.0223	.0456	.0936	.0909	−.0755	−.0779	−.0258	.0147	−.0515	−.0320	.0655	37.24

NOTE.—YP = yearly growth rate in industrial production; MP = monthly growth rate in industrial production; DEI = change in expected inflation; UI = unanticipated inflation; UPR = unanticipated change in the risk premium (Baa and under return − long-term government bond return); UTS = unanticipated change in the term structure (long-term government bond return − Treasury-bill rate); CG = growth rate in real per capita consumption; OG = growth rate in oil prices; EWNY = return on the equally weighted NYSE index; and VWNY = return on the value-weighted NYSE index.

5.3 Table 4 and Table 5: macro variables versus market indices

Read: Table 4:

- Part A includes YP, MP, DEI, UI, UPR, and UTS.
- In the full 1958-84 period, MP is strongly significant with a coefficient of 13.984 and a t-statistic of 3.727.
- UI is negative and significant in the full period, and the inflation variables are especially strong in 1968-77.
- UPR is positive and significant, with a full-period coefficient of 7.941 and a t-statistic of 2.807.
- UTS is negative and marginally significant in the full period.
- Parts C and D add the equally weighted and value-weighted NYSE indices. The market-index variables are not significant, while the macro variables remain important.

Read: Table 5:

- Part A gives a pure market-index test: the VWNY beta is significant over the whole sample.
- Part B adds macro variables. The VWNY beta becomes negative, while MP, UI, and UPR remain significant.
- Part C estimates the VWNY beta over the whole testing period. The market-index beta becomes insignificant, while MP, DEI, UI, UPR, and UTS remain significant in the full period.

Economic message. These are the paper's central tables. The market index explains time-series comovement, but macroeconomic exposures explain cross-sectional expected returns better.

TABLE 4 **Economic Variables and Pricing (Percent per Month × 10), Multivariate Approach**

A

Years	YP	MP	DEI	UI	UPR	UTS	Constant
1958–84	4.341	13.984	−.111	−.672	7.941	−5.87	4.112
	(.538)	(3.727)	(−1.499)	(−2.052)	(2.807)	(−1.844)	(1.334)
1958–67	.417	15.760	.014	−.133	5.584	.535	4.868
	(.032)	(2.270)	(.191)	(−.259)	(1.923)	(.240)	(1.156)
1968–77	1.819	15.645	−.264	−1.420	14.352	−14.329	−2.544
	(.145)	(2.504)	(−3.397)	(−3.470)	(3.161)	(−2.672)	(−.464)
1978–84	13.549	8.937	−.070	−.373	2.150	−2.941	12.541
	(.774)	(1.602)	(−.289)	(−.442)	(.279)	(−.327)	(1.911)

B

Years	MP	DEI	UI	UPR	UTS	Constant
1958–84	13.589	−.125	−.629	7.205	−5.211	4.124
	(3.561)	(−1.640)	(−1.979)	(2.590)	(−1.690)	(1.361)
1958–67	13.155	.006	−.191	5.560	−.008	4.989
	(1.897)	(.092)	(−.382)	(1.935)	(−.004)	(1.271)
1968–77	16.966	−.245	−1.353	12.717	−13.142	−1.889
	(2.638)	(−3.215)	(−3.320)	(2.852)	(−2.554)	(−.334)
1978–84	9.383	−.140	−.221	1.679	−1.312	11.477
	(1.588)	(−.552)	(−.274)	(.221)	(−.149)	(1.747)

C

Years	EWNY	MP	DEI	UI	UPR	UTS	Constant
1958–84	5.021	14.009	−.128	−.848	8.130	−5.017	6.409
	(1.218)	(3.774)	(−1.666)	(−2.541)	(2.855)	(−1.576)	(1.848)
1958–67	6.575	14.936	−.005	−.279	5.747	−.146	7.349
	(1.199)	(2.336)	(−.060)	(−.558)	(2.070)	(−.067)	(1.591)
1968–77	2.334	17.593	−.248	−1.501	12.512	−9.904	3.542
	(.283)	(2.715)	(−3.039)	(−3.366)	(2.758)	(−2.015)	(.558)
1978–84	6.638	7.563	−.132	−.729	5.273	−4.993	9.164
	(.906)	(1.253)	(−.529)	(−.847)	(.663)	(−.520)	(1.245)

D

Years	VWNY	MP	DEI	UI	UPR	UTS	Constant
1958–84	−2.403	11.756	−.123	−.795	8.274	−5.905	10.713
	(−.633)	(3.054)	(−1.600)	(−2.376)	(2.972)	(−1.879)	(2.755)
1958–67	1.359	12.394	.005	−.209	5.204	−.086	9.527
	(.277)	(1.789)	(.064)	(−.415)	(1.815)	(−.040)	(1.984)
1968–77	−5.269	13.466	−.255	−1.421	12.897	−11.708	8.582
	(−.717)	(2.038)	(−3.237)	(−3.106)	(2.955)	(−2.299)	(1.167)
1978–84	−3.683	8.402	−.116	−.739	6.056	−5.928	15.452
	(−.491)	(1.432)	(−.458)	(−.869)	(.782)	(−.644)	(1.867)

TABLE 5 **Economic Variables and Pricing**

A

Years	VWNY	MP	DEI	UI	UPR	UTS	Constant
1958–84	14.527	...	...	...	...	...	−5.831
	(2.356)						(−.961)
1958–67	5.005	...	...	...	...	...	6.853
	(.673)						(.928)
1968–77	17.987	...	...	...	...	...	−15.034
	(1.460)						(−1.254)
1978–84	23.187	...	...	...	...	...	−10.802
	(1.935)						(−.907)

B

Years	VWNY	MP	DEI	UI	UPR	UTS	Constant
1958–84	−9.989	12.185	−.145	−.912	9.812	−5.448	10.714
	(−2.014)	(3.153)	(−1.817)	(−2.590)	(3.355)	(−1.609)	(2.755)
1958–67	−5.714	13.024	.004	−.193	6.104	−.593	9.527
	(−1.008)	(1.852)	(.057)	(−.369)	(1.994)	(−.260)	(1.983)
1968–77	−17.396	14.467	−.291	−1.614	14.367	−9.227	8.584
	(−1.824)	(2.214)	(−3.388)	(−3.297)	(3.128)	(−1.775)	(1.167)
1978–84	−5.515	7.725	−.150	−.935	8.602	−6.986	15.454
	(−.513)	(1.303)	(−.574)	(−1.051)	(1.064)	(−.681)	(1.867)

C

Years	VWNY	MP	DEI	UI	UPR	UTS	Constant
1958–84	11.507	10.487	−.190	−.738	8.126	−7.073	−3.781
	(1.189)	(2.761)	(−2.459)	(−2.215)	(2.869)	(−2.194)	(−.402)
1958–67	22.311	9.597	.001	−.163	3.186	.697	−11.734
	(1.950)	(1.494)	(.012)	(−.341)	(1.474)	(.337)	(−1.015)
1968–77	11.689	13.381	−.293	−1.422	13.007	−12.981	−9.488
	(.622)	(1.947)	(−3.590)	(−2.814)	(2.697)	(−2.214)	(−.526)
1978–84	−4.188	7.624	−.316	−.584	8.211	−9.735	15.732
	(−.207)	(1.286)	(−1.246)	(−.716)	(1.039)	(−1.123)	(.803)

5.4 Table 6 and Table 7: consumption and oil alternatives

Read: Table 6:

- The consumption-growth beta is not significant in the full 1964-84 period, with a t-statistic of only 0.108.
- Consumption is also not significant in either subperiod.
- The macro variables, especially MP, UPR, and UTS, remain meaningful when consumption is included.

Read: Table 7:

- Oil price growth is not significant in the full sample.
- Oil is not priced in the crucial 1968-77 subperiod when OPEC became important.
- Adding oil reduces the significance of industrial production but strengthens the significance of UPR and UTS.

Economic message. Consumption and oil are natural alternatives, but neither displaces the macro state variables. This reinforces the paper's conclusion that industrial production, inflation, risk-premium, and term-structure news are the important priced forces.

TABLE 6 **Pricing with Consumption**

Years	CG	MP	DEI	UI	UPR	UTS	Constant
1964–84	.68	14.964	−.166	−.846	8.813	−6.921	2.289
	(.108)	(3.800)	(−1.741)	(−2.250)	(2.584)	(−1.790)	(.628)
1964–77	−.485	18.150	.166	−.946	11.442	−9.191	−1.910
	(−.659)	(3.535)	(−2.419)	(−2.494)	(3.288)	(−2.412)	(−.442)
1978–84	1.173	8.592	−.166	−.645	3.556	−2.382	10.687

Table 6: Pricing with consumption

TABLE 7 **Pricing with Oil Price Changes**

Years	OG	MP	DEI	UI	UPR	UTS	Constant
1958–84	2.930	12.728	−.095	−.391	11.844	−8.726	4.300
	(.996)	(1.406)	(−1.193)	(−1.123)	(4.294)	(−2.770)	(1.340)
1958–67	4.955	14.409	.078	.119	8.002	−1.022	2.663
	(1.978)	(.921)	(1.102)	(.204)	(2.604)	(−.421)	(.556)
1968–77	1.038	4.056	−.223	−1.269	16.170	−16.055	−1.344
	(.251)	(.296)	(−2.737)	(−2.975)	(3.839)	(−3.154)	(−.243)
1978–84	2.738	22.718	−.159	.134	11.152	−9.264	14.702
	(.303)	(1.228)	(−.598)	(.156)	(1.465)	(−1.024)	(2.240)

NOTE.—CG = growth rate in real per capita consumption; OG = growth rate in oil prices; VWNY = return on the value-weighted NYSE index; EWNY = return on the equally weighted NYSE index; MP = monthly growth rate in industrial production; DEI = change in expected inflation; UI = unanticipated inflation; UPR = unanticipated change in the risk premium (Baa and under return −

Table 7: Pricing with oil price changes

6 Short summary

- Chen, Roll, and Ross ask which macroeconomic forces are priced in stock returns.
- They start from the valuation identity that prices depend on expected cash flows and discount rates.
- They construct macro variables that proxy for cash-flow and discount-rate news.
- The main variables are industrial production growth, unexpected inflation, changes in expected inflation, the default-spread/risk-premium variable, and the term-structure variable.
- They use 20 size-sorted stock portfolios and a Fama-MacBeth procedure.
- The main pricing period starts in 1958 because the paper estimates betas using a prior 60-month window.
- Industrial production is strongly priced in the full sample.
- The risk-premium variable UPR is positively priced.
- The term-structure variable UTS is negatively priced.
- Inflation variables matter especially in the high-inflation 1968-77 period.
- Market indices are powerful time-series variables but weak cross-sectional pricing variables once macro betas are included.
- Consumption growth is not significantly priced.
- Oil price growth is not independently priced.
- The paper's broad contribution is to give economic labels to priced systematic risks rather than relying only on statistical factors or the market portfolio.

7 Conclusion

7.1 Core conclusion

Stock returns are exposed to systematic economic news, and expected returns are related to those exposures. Industrial production, inflation-related variables, risk-premium shifts, and term-structure shifts are priced in the cross-section of stock returns. Market indices, consumption growth, and oil price growth do not displace these variables.

7.2 Economic intuition

The economic logic is simple: stock values are discounted expected cash flows. Production news changes expected real cash flows. Inflation news changes nominal cash flows, interest rates, and wealth distribution across assets. Risk-premium shocks change the compensation investors require for bearing uncertainty. Term-structure shifts change the value of long-duration claims. Stocks that covary differently with these forces should command different expected returns.

7.3 Takeaway

The paper is an early and influential attempt to identify macroeconomic state variables behind systematic risk. Its message is that the market portfolio is not the only useful pricing object. Economically interpretable macro variables can explain why some portfolios earn higher average returns than others.